

Pre-feasibility study to establish a Dolomites factory



2017



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1. Exclusive summary

Dolomite ores are occurs in Tafila, which is located at south of the kingdom, dolomite is of high economic feasibility as raw materials, where the reserves in the area between Assal valley, Ahimr valley estimated 62 million tons and reserves of Hadeeth Valley area estimated as 20 million tons. Due to the presence of these abundant quantities of ores, the idea of setting up a factory manufacturing the raw dolomite ore economic will be feasible to Kingdom economy

Table1 : Project Main Indicators

Indicator	Value
Total Investment Cost (Jordan Dinars)	2.269.723
Internal Rate of Return (IRR)	%23.38
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	1.895.301
Job Opportunities provided by the Project	15
Total Revenues (For 10 Years)	15.820.153
Total Operating Expenses (For 10 years)	8.440.052
Total Net Profit (For 10 years)	2.140.175

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

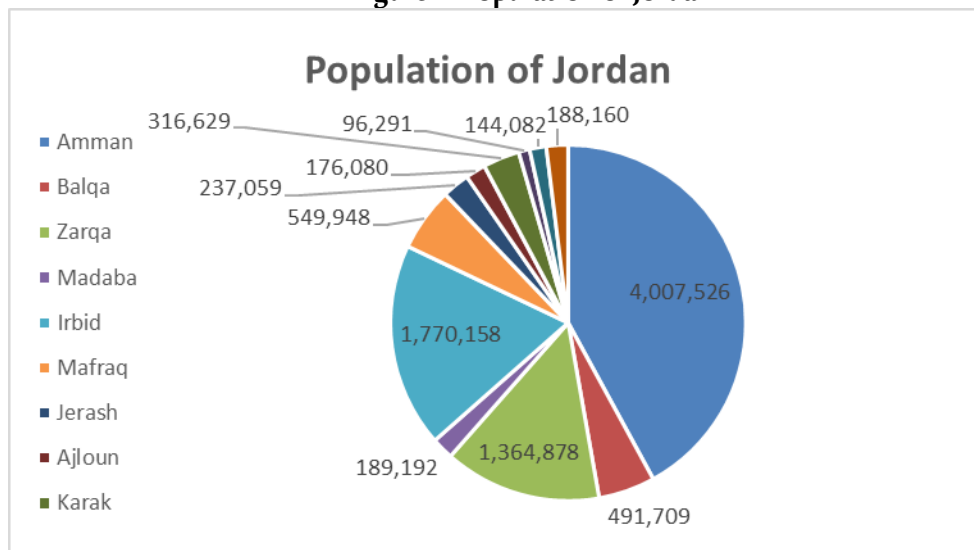
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

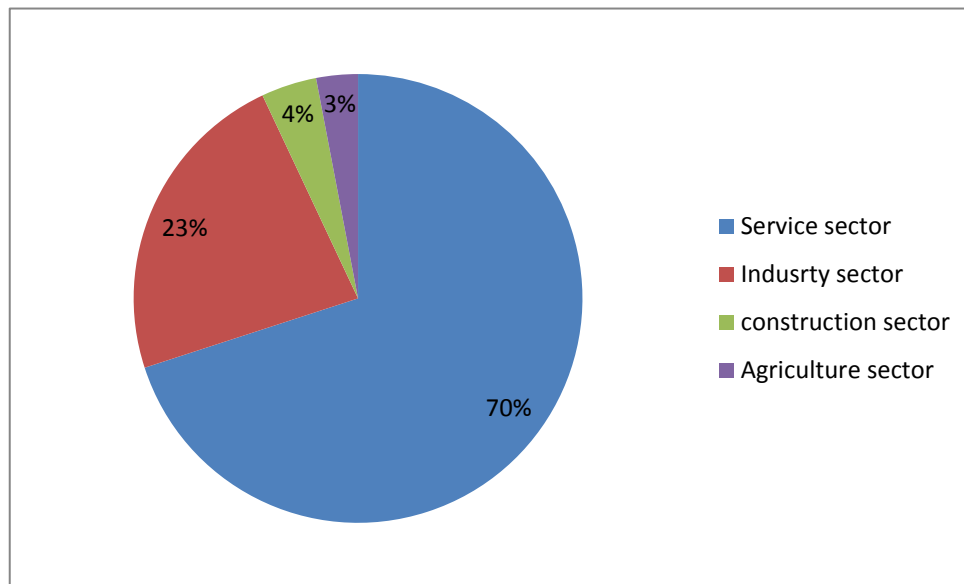
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy

facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Geographic and demographic reality

Tafila Jordan is one of the provinces in the western sector of the Kingdom. Bordered to the north of Karak and the east and south of Ma'an and Aqaba governorate. One of the most important and most spectacular reserves the Middle East there is a Dana. And away from the capital, Amman, about 180 km. The Tafeileh of the oldest populated areas, where successively by different nations: Starting Baladumien then the area subject to the rule of the Nabataeans that the Romans came to the area and while its former name was de Twlos means or chrome then underwent Islamic rule after the Battle of Mutah and the Battle of Yarmuk. 2016 total population Tafeileh 96,291 people - (Department of Statistics).

Natural Resources: The Tafeileh are distinct natural resources but has not been exploited fully and the most important metals in which copper is estimated reserves of pure metal known about a million metric tons. The manganese, known reserves of it is estimated at about half a million metric tons

Tafila is rich in minerals and industrial rocks, which called for a number of senior Jordanian companies to exploit some of these resources, such as the Jordan Phosphate Mines Company, which makes phosphate ores in Hasa region, and Jordan Cement Factories Company for the production of cement in Rashadiya factory. It also contains limestone, clay and industrial rocks available in the province are in the shale cliff dervishes' area, perlite near Hasa, and dolomite near Rashadiya, and gypsum in a mountain area near Melih Tafeileh

The investment environment

The mining sector:

The Jordan Phosphate Mines Company exploits the available phosphate ores in Hasa region, and as the Jordan Cement Factories Company exploit the raw materials needed to produce cement in Rashadiya factory and the most important of these raw materials are limestone, clay and clay

investments amounted made to take advantage of Investment Promotion Law in Tafila for the years Law Size (2013-2014) to \$ (792.8) million, of which (192.0) million JD for 2014, compared with the total volume of investments for the same period, the value of which was estimated at (1.2456) million JD. The number of investment opportunities for the province as one of the outputs of Tafila investment map of the Kingdom two investment opportunities with a total value of approximately 117 million JD, and jobs unexpected (175) opportunity. * 2015 Investment Authority

The most prominent investment opportunities in Tafila

- exploit uranium to produce atomic energy
- Building Materials Industries (gypsum, zeolite)
- fertilizer industry and chemical industry
- the exploitation of oil shale to generate electricity

* Source: Study of the investment map for the southern provinces. Arab Planning Institute in 2015.

Ease of Doing Business

- simplified procedures for registration and licensing of various economic activities and through a unified investment window. Where he set up a company or investor can become the institution through a representative of the Ministry of Industry and Trade, who is in Investor Services Directorate and to register the company or organization to take advantage of the benefits of the law. As well as to learn about the permit requirements of direct action and public safety, public health and environmental requirements to be met, and begin the process of obtaining the necessary permits and certificates and approvals. This window also allows the investor to apply for land and work permits and residency for foreign workers needed.
- There is no restriction on the proportion of foreign investment in the contribution of all tourism, industrial, agricultural and commercial areas and most of the services with the exception of some professional services.
- There is no restriction on the transfer of profits and capital abroad
- There is no restriction on dealing in foreign currencies.

Project description

Given the economic importance of mining projects that provide many advantages such as creating jobs and opening export prospects, the exploitation of raw dolomite through extraction and manufacturing to be used in several fields such as the tile, industry and its use, in improving agricultural soil and many other uses is cost-effective because of the importance of this raw and its chemical and industrial composition.

Dolomite is a complex mixture of calcium carbonate and magnesium and chemical composition $\text{Ca, Mg} (\text{CO}_3)_2$ dolomite called of the metal as well as the carbonate rocks that contains more than 50% of the dolomite mineral. Dolomitic rocks is divided according to its metal contents of dolomite and calcite into the following types: Alcalceta dolomite, limestone dolomite, magnesium dolomite.

Pure dolomite theoritically consists of:

- Calcium oxide 30.4%
- Maganisum oxide 21.3%
- Carbon dioxide 47.8%

Properties and existence in nature:

dolomite in nature are often occurred in graded colors from white to gray through brown, red or pink or black, has a pearly luster or glass, and it exists in the form of curved crystals. And it is often identical with limestone rocks, and there is also the rock Alserbntinett and some other Almgnesah rocks. It consists of dolomite as a result of the process of turning rock limestone dolomitization or secondary configuration is a common way to spread. And also directly formed by precipitation of seawater; a chemical method, this kind known as a primary origination dolomite which is rarely occurred.

Uses of the Dolomites:

Dolomite is used in many diffrient fields such as:

- Aggregate Production that is necessary for road works and reinforced concrete
- the production of ornamental stones, specifcly amorphous dolomite
- production of breaking dolomite necessary for the manufacturing of tiles and limestone mosaic bricks
- Agriculture: to improve the soil and rebalance-acidity

- Refractories: for lining furnaces and molds used in steel production
- Catalyst materials: to reduce the melting point of some metals and increase liquidity
- Glass: as catalyst and to increase the strength of the final product
- Production of insulating materials by adding dolomite to clay fiery
- as a filler in the production of paints, plastics and paper
- the production of magnesia: the Dolomites is used in the magnesium and calcium production, by separating them from each other its can be used in many industries, the magnesium oxide is used in the magnetic iron sheets industry, and most uses of the magnesium oxides is in the pharmaceutical industry as well as, in the cosmetics industry. The magnesium carbonate and calcium carbonate are both used as a base for the manufacturing of creams and toothpastes.

Table 2 physical properties of dolomite

Chemical composition	Ca, Mg(CO ₃) ₂
Crystal family	triclinic
colour	Transparent usually white
Scratch colour	white
Sparkle	glass
Transparency	Transparent to translucent
hardness	4-3.5

Table 3 chemical properties

Area	%MgO	%SiO ₂	%CaO	%Fe ₂ O ₃
Wadi assal	4,77-18,98	0,95-6,44	31,13-46,7	0,12-1,36
Ghaur hadetha	1,74-20,2	0,45-24,2	21,55-50,9	0,1-3,57

The Project Objectives

- Exploitation of Jordanian ores
- Establish a factory to exploit and manufacturing the dolomite ores manufacturing
- encourage investors to enter industries which rely on this raw material
- Open the prospects for exports outside Jordan
- Create jobs and to employ the qualified Jordanians.

Targeted categories

There are many target groups concrete admixtures factories are the most important such as:

1. Almanaseer for concret
2. Lafarj factory
3. Others
4. Paints Factories such as:
 1. Delux
 2. Remalux
 3. National
 4. Others

Analysis of markets

Table 4 Reserves of dolomite ore in Jordan

Area	Reserve(million/ton meter)
Area between wadi assal and wadi ehamar	62
Ghour alhadetha	20
Total	82

Source: Energy and Minerals Regulatory Commision.

Competitor analysis

Till now, there is no investment for dolomite, but now it is open for companies and investors to exploit raw dolomite ore and open export prospects

Market Share

Due to unavailability of sources showing the amount of import and local use, it is assumed that the exploitation of these raw materials will be based on the practical and technical experience, as well as, producing these raw materials according to industrial specifications to be used in the local industries in addition to export.

- It is definitely feasible to exploit these raw materials and extract it by establishing a factory producing 100,000 tons / year, 333.3 tons / day 8 hours. According to the chemical analyzes of the sites the the products after resulting from the manufacturing process according to as magnesium oxide ratios and calcium oxide listed in the table (2) are
- Calcium oxide dolomite
- Maganisiium oxide dolomite
- Iron

Price analysis and pricing policy

The dolomite prices which have added value is vary according to the type of treatment and useage. For example, not limited to the price per ton of magnesium component differs from the price ton of chemicals magnisum and the price per ton of treated thermally types suitable for metallurgical and chemical industries. As for the price refined and grinded ore necessary for the production of fertilizers and glass per ton of ranges between 11-18 JD, price per ton for raw materials used in the production of aggregates 2-4 JD - base on the international price of the Dolomites.

Exporters and importers

Global exports about 10 million ton, the most important exporting countries are Canada, Belgium, Slovakia, China and Thailand. The main importing countries are the United States, Japan, Netherlands, Germany, France, Chile, where, there is no import or dolomites ore.

Table 5 prices assumptions

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Industrial Dolomite.	15	15	16	16	17	17	18	18	19	20
Agregate Dolomite	3	3	3	3	3	3	4	4	4	4

Expected Revenues

Table 6 expected revenues of project during the years 2018-2027

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Industrial Dolomite revenues	1350000	1390500	1432215	1475181	1519437	1565020	1611971	1660330	1710140	1761444
Agregate Dolomite revenues	30000	30900	31827	32782	33765	34778	35822	36896	38003	39143
Total Revenues	1380000	1421400	1464042	1507963	1553202	1599798	1647792	1697226	1748143	1800587

2. Technical analysis

Human resources required

The following table shows the required human resources for the project:

Table7 Human resources required for the project

position	number	sallary
Manager project	1	1,500
Geologist	1	550
Accountant	1	400
Resptionist	1	300
Technician	2	350
Maintenance	1	350
Cleaninig labour	1	250
Driver	5	380
Security gaurd	2	300

- **Project Manager:** supervise the works of the staff, and do make decisions related to the production process and the process of distribution and deployment of the product in the local market. Insure the availability of safety requirements and occupantional health in production sites.
- **Geologist:** apply the theoretical knowledge and research data available to determine the mineral resources and potential future development sites
- **Accountant:** Follow up aspects material and the financial audit.
- **Maintenance:** Insure carrying out the procedures and activities of the electrical and mechanical maintenance, and follow up the implementation of plans and programs planned in order to ensure the continuity of the production process and quality at full capacity and less expenses as possible, and follow up availability of spare parts required, and the preparation of technical reports.
- **Cleaning labour:** To maintain the cleanliness of the plant and offices.
- **Driver:** To transfer the products to customers.

Location analysis

The most important sites of dolomite ore occurrences:

The area between the Valley and Wadi Assal Ahimrasal located, about 30 km to the west of the city of Karak and to the east of the road that connected the Ghor Safi-aqaba

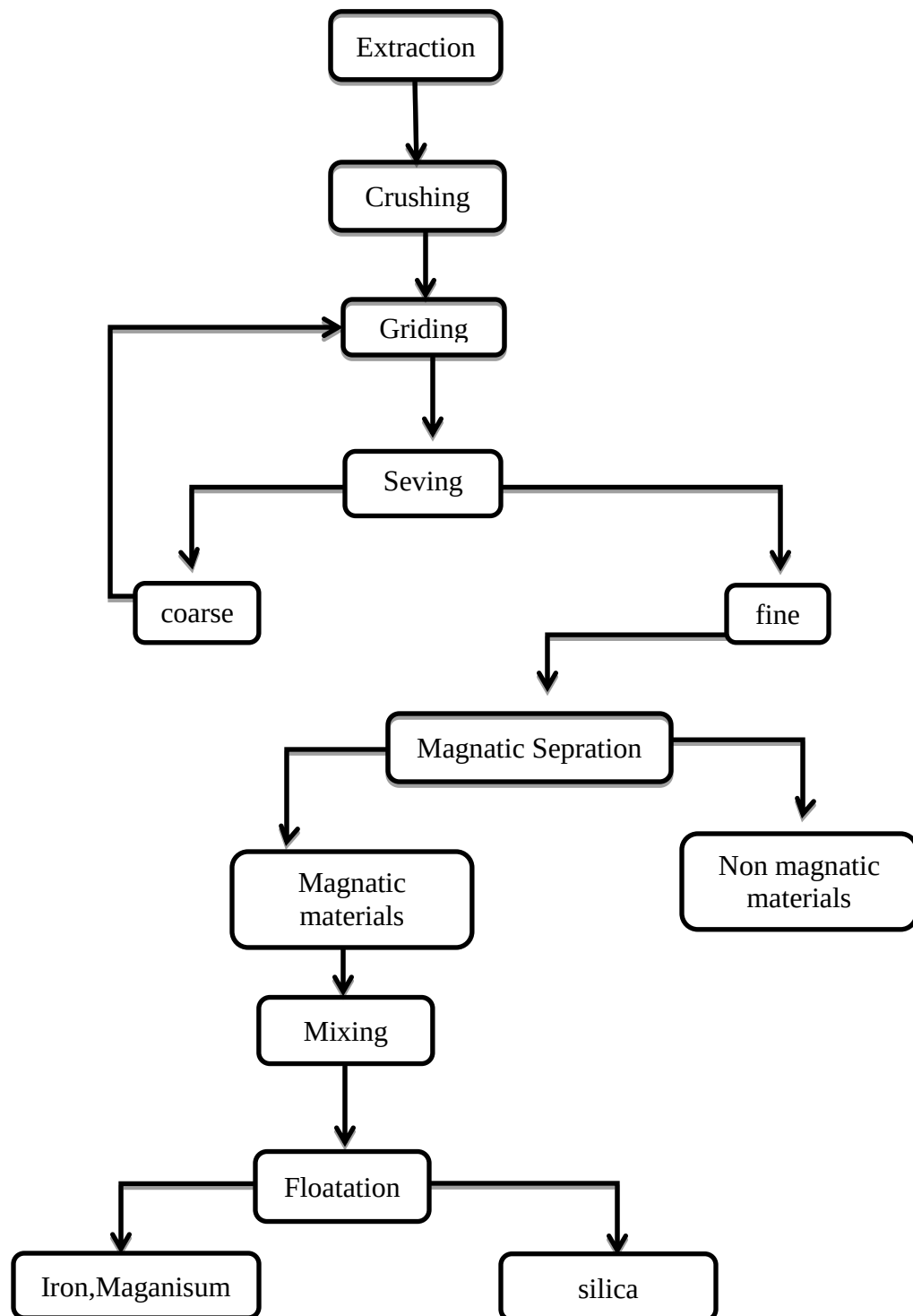
Ghour alhadetha area located, about 25 km to the west of the city of Karak and to the east of Gore al hadetha town.

After reviewing the table (4) and the geographical location in figure (1) below, the area between Wadi Assal and Wadi Ahimr taking into account its proximity to the highway and ore occurrences with reserves of a 62 million / mt can be considered the best area for the purposed project.

Figure (1): ore occurrences sites dolomite-source the energy and minerals sector regularity commission.



Description of production process and steps of manufacturing



Dolomite ores are extracted from the sites and then start the processing, the first followed by milling and grinding machine conical milling process and is then separated by raw gradation using Sieves to be re rough part for milling and part soft enters the magnetic separation processes where the ratio of magnetic materials is at 70% others and the other subject to magnetize that are mixed and then introduced into the flotation processes to get the Celica 10%, iron and Aghannaseyoum by 20%.

Technical requirement analysis

Earth and construction works

It was assumed that a piece of land of 30 dunom will be leased from the concerned at department governorate after carrying out the exploration covering an area of 1 km 2. And Table 8 below shows the fees required for the project according to the recommendations of the Central Committee of the state property / Energy and Mineral sector Regulator Commision.

Table8 mining fees

Requirements	Mining fees
Feesa prospecting licencer km2	200
Fees of mining rights granted annually on km2	500
Fees prospecting permits site visit	500
Fees Treasury land for prospecting km2	800
Fees of annual rent 25 JD per dunam, 30 dunum	750
Total	2,750

Equipments and machines

The table (9) below shows the equipments and machines required for the project:

Table 9 the equipments and machines required for the project

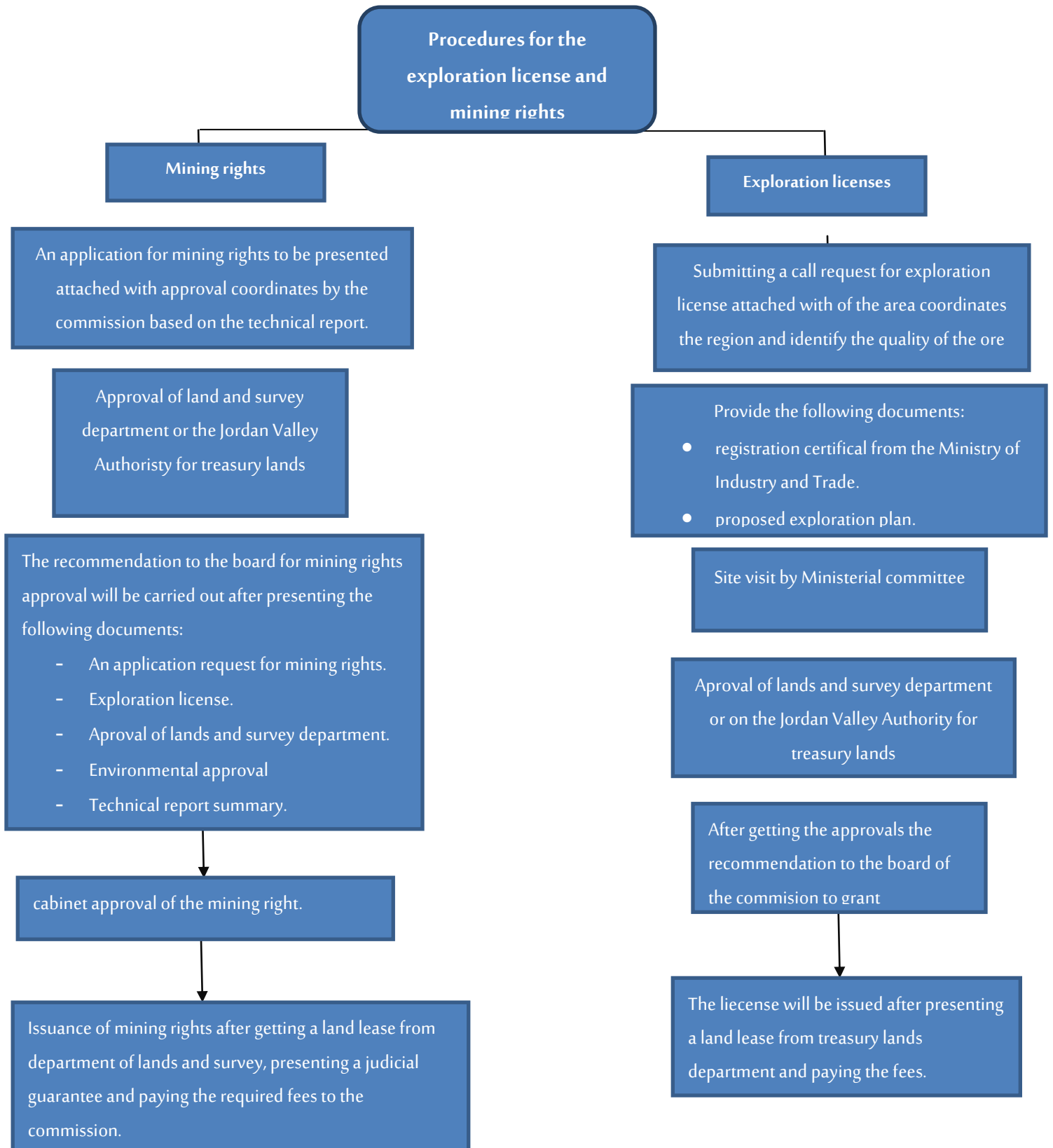
Equipments and machine	number	Cost(JD)/unit
loder	2	60,000
Crushing machine	1	40,000
Floatation machine	1	40,000
Magnatic machine	1	355,000
Sieving machine	2	25,000
griding machine	1	70,000
dumber	2	400,000
mixer	1	50,000

Furnitures

Office furnitures (computers, tables and chaires, carpet, meeting requierments) that cost 12,000 JD

Requirments and legal procedures

The following figure shows the necessary procedures for exploration and mining rights license depending on body regulating energy and minerals sectors:



4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 20 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 2 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 3 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 4 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 5 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	40.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table 6 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	5.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table7 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	907,889
Loans Value	1,361,834
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	1,361,834
Loans Percentage	%60

Equity	
Cost of Equity	%15.5
Equity Value	907,889
Equity Percentage	%40
Gross	
Project Value	2,269,723
Weighted Average Cost of Capital	%10.75

Capital Expenditures

The estimated cost of the project is 2,269,723 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 8 Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross Cost
Construction Works	200	200	40,000
Hungers	200	1,000	200,000
Paving	15	500	7,500
Other Works			12,375
Total			259,875

Table 9 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
loder	2	60,000	120,000
Crushing machine	1	40,000	40,000
Floatation machine	1	40,000	40,000
Magnatic machine	1	355,000	355,000
Sieving machine	2	25,000	50,000
Griding machine	1	70,000	70,000
dumber	1	400,000	400,000
mixer	1	50,000	50,000
dumber	1	400,000	400,000
Total			1,525,000

Table : 10 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table11 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	184,058
General and administrative expenses	3	%25	6,890
Marketing expenses	3	%25	863
Indirect Salaries	3	%25	26,200
Total			218,010

Table 12 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	259,875	10%	285,863	%12.6
Machinery & Equipment	1,525,000	10%	1,677,500	%73.9
Furniture	12,000	10%	13,200	%0.6
Vehicles	46,000	10%	50,600	%2.2
Pre-Operating Expenses	2,500	10%	2,750	%0.1
Working capital	218,010	10%	239,811	%10.6
Total (JD)	2,063,385		2,269,723	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,269,723) JD, where 60% of the project will be financed through loans (1,361,834) JD, and the remaining (40%) through (Equity) with (907,889) JD. The following table summarizes the general structure of the required financing:

Table 13 Sources of Funding

Funding Sources	Amount	Percentage
Equity	907,889	%40
Loans	1,361,834	%60
Total	2,269,723	%100
Use of Fund:	Value	Ratio
Capital Expenditures	2,027,163	%89.31
Pre-Operating Expenses	2,750	%0.12
Working Capital	239,811	%10.57
Total	2,269,723	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 14 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	552,000	568,560	585,617	603,185	621,281	639,919	659,117	678,890	699,257	720,235
Utilities from Revenues	69,000	71,070	73,202	75,398	77,660	79,990	82,390	84,861	87,407	90,029
Maintenance from Revenues	34,500	35,535	36,601	37,699	38,830	39,995	41,195	42,431	43,704	45,015
Disposables from Revenues	13,800	14,214	14,640	15,080	15,532	15,998	16,478	16,972	17,481	18,006
Others	66,930	68,938	71,006	73,136	75,330	77,590	79,918	82,315	84,785	87,328
Total	736,230	758,317	781,066	804,498	828,633	853,492	879,097	905,470	932,634	960,613

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 15 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	15,204	15,660	16,130	16,614	17,112	17,625	18,154	18,699	19,260	19,837
Miscellaneous	2,505	2,581	2,658	2,738	2,820	2,904	2,992	3,081	3,174	3,269
Total	27,559	28,386	29,237	30,115	31,018	31,949	32,907	33,894	34,911	35,958

Marketing Expenses

Table 16 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	3,450	3,554	3,660	3,770	3,883	3,999	4,119	4,243	4,370	4,501

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table :17 Expected Annual Salaries and Wges

Job title	No.	Monthly salary (JD)	Total monthly salaries	The total annual salaries (JD)
Manager project	1	1,500	1,500	24,000
geology	1	550	550	8,800
accounter	1	400	400	6,400
resption	1	300	300	4,800
Technical mining	2	350	700	11,200
maintenance	1	350	350	5,600
Cleaninig man	1	250	250	4,000
driver	5	380	1,900	30,400
Security gaurd	2	300	600	9,600
Total	15		6,550	104,800

Table : 18 Expected Annual Salaries and Wges (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Manager project	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
geology	8,800	9,161	9,536	9,927	10,334	10,758	11,199	11,658	12,136	12,634
accounter	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
resption	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Technical mining	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
maintenance	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Cleaninig man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
driver	30,400	31,646	32,944	34,295	35,701	37,164	38,688	40,274	41,926	43,645
Security gaurd	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Total	104,800	109,097	113,570	118,226	123,073	128,119	133,372	138,841	144,533	150,459

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 19 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	285,863	%5.0	%0.0
Machinaries	1,677,500	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0

Table 20 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	2,027,163	2,061,373	2,096,287	2,131,920	2,168,287	2,205,405	2,243,289	2,281,956	2,321,424	2,361,708
Total Depreciation Expenses	188,423	191,844	195,336	198,899	202,536	206,247	210,036	213,903	217,849	221,878
Total Accumulated Depreciation	188,423	380,267	575,603	774,502	977,037	1,183,285	1,393,320	1,607,223	1,825,072	2,046,950
Total Additions	0	34,210	34,914	35,633	36,368	37,118	37,884	38,667	39,467	40,284
Total Net Book Values	1,838,739	1,681,105	1,520,684	1,357,418	1,191,250	1,022,120	849,969	674,734	496,351	314,758

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table : 21 Loan Details

Loan Value	1,361,834
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	354,489
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,361,834
2019	354,489	129,102	225,387	1,136,447
2020	354,489	107,735	246,753	889,694
2021	354,489	84,343	270,146	619,548
2022	354,489	58,733	295,755	323,793
2023	354,489	30,696	323,793	0

Income Statement

Table 22 Income Statement

الوصف	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,380,000	1,421,400	1,464,042	1,507,963	1,553,202	1,599,798	1,647,792	1,697,226	1,748,143	1,800,587
Gross Operating Revenues	1,380,000	1,421,400	1,464,042	1,507,963	1,553,202	1,599,798	1,647,792	1,697,226	1,748,143	1,800,587
Operating Expenses	-736,230	-758,317	-781,066	-804,498	-828,633	-853,492	-879,097	-905,470	-932,634	-960,613
Gross Operating Profit	643,770	663,083	682,976	703,465	724,569	746,306	768,695	791,756	815,509	839,974
<i>Gross Profit Percentage</i>	%47	%47	%47	%47	%47	%47	%47	%47	%47	%47
Salaries and Benefits (Indirect Staff)	-104,800	-109,097	-113,570	-118,226	-123,073	-128,119	-133,372	-138,841	-144,533	-150,459
General and Administraive Expenses	-27,559	-28,386	-29,237	-30,115	-31,018	-31,949	-32,907	-33,894	-34,911	-35,958
Markeing Expenses	-3,450	-3,554	-3,660	-3,770	-3,883	-3,999	-4,119	-4,243	-4,370	-4,501
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-138,559	-141,036	-146,467	-152,111	-157,974	-164,067	-170,399	-176,978	-183,814	-190,919
Income before Interest, Depreciation and Tax	505,211	522,047	536,508	551,354	566,594	582,238	598,296	614,778	631,694	649,055
Fixed Assets Depreciations	-188,423	-191,844	-195,336	-198,899	-202,536	-206,247	-210,036	-213,903	-217,849	-221,878
Income before Tax and Interests	316,788	330,203	341,173	352,455	364,059	375,991	388,260	400,876	413,845	427,177
Bank Interests	-129,102	-107,735	-84,343	-58,733	-30,696	-0	-0	-0	-0	-0
Income Before Tax	187,686	222,468	256,830	293,722	333,363	375,991	388,260	400,876	413,845	427,177

الوصف	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Tax	-87,537	-94,494	-101,366	-108,744	-116,673	-125,198	-127,652	-130,175	-132,769	-135,435
Net Profit	100,149	127,974	155,464	184,978	216,691	250,793	260,608	270,700	281,076	291,742
<i>Net Profit Percentage</i>	<i>%7</i>	<i>%9</i>	<i>%11</i>	<i>%12</i>	<i>%14</i>	<i>%16</i>	<i>%16</i>	<i>%16</i>	<i>%16</i>	<i>%16</i>
Compulsory Reserves	-10,015	-12,797	-15,546	-18,498	-21,669	-25,079	-26,061	-27,070	-28,108	-29,174
Retained Earnings	90,134	205,311	345,228	511,708	706,730	932,443	1,166,991	1,410,621	1,663,589	1,926,157

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 23 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	100,149	127,974	155,464	184,978	216,691	250,793	260,608	270,700	281,076	291,742
Bank Interests	129,102	107,735	84,343	58,733	30,696	0	0	0	0	0
Depreciation	188,423	191,844	195,336	198,899	202,536	206,247	210,036	213,903	217,849	221,878
Total Operating Cashflows before Additions to Working Capital	417,674	427,553	435,142	442,610	449,922	457,040	470,644	484,603	498,925	513,620
Inventory (Increase/Decrease)	-46,000	-1,380	-1,421	-1,464	-1,508	-1,553	-1,600	-1,648	-1,697	-1,748
Accounts Receivables (Increase/Decrease)	30,705	921	949	977	1,007	1,037	1,068	1,100	1,133	1,167
Working Capital (Increase/Decrease)	-15,295	-459	-473	-487	-501	-516	-532	-548	-564	-581
Net Cashflows from Operating Activities	402,379	427,095	434,670	442,123	449,420	456,524	470,112	484,055	498,361	513,038
Cashflows from Investments Activities										
Fixed Assets (Procurement)	-2,027,163	-34,210	-34,914	-35,633	-36,368	-37,118	-37,884	-38,667	-39,467	-40,284
Net Cashflows from Investments Activities	-2,027,163	-34,210	-34,914	-35,633	-36,368	-37,118	-37,884	-38,667	-39,467	-40,284
Cashflows from Financing Activities										
Capital	907,889									

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Loan Amortization	-225,387	-246,753	-270,146	-295,755	-323,793	0	0	0	0	0
Bank Interest Rate	-129,102	-107,735	-84,343	-58,733	-30,696	-0	-0	-0	-0	-0
Loans	1,361,834	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,915,235	-354,489	-354,489	-354,489	-354,489	0	0	0	0	0
Net (Increase/Decrease) in Cash	290,451	38,396	45,267	52,001	58,564	419,406	432,228	445,388	458,894	472,754
Cashflows at the Beginning of Period	0	290,451	328,847	374,114	426,115	484,680	904,086	1,336,314	1,781,702	2,240,595
Cashflows at the End of Period	290,451	328,847	374,114	426,115	484,680	904,086	1,336,314	1,781,702	2,240,595	2,713,349

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 24 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	290,451	328,847	374,114	426,115	484,680	904,086	1,336,314	1,781,702	2,240,595	2,713,349
Inventory	46,000	47,380	48,801	50,265	51,773	53,327	54,926	56,574	58,271	60,020
Total current Assets	336,451	376,227	422,915	476,381	536,453	957,412	1,391,240	1,838,276	2,298,867	2,773,369
Non Current Assets										
Fixed Assets (net)	1,838,739	1,681,105	1,520,684	1,357,418	1,191,250	1,022,120	849,969	674,734	496,351	314,758
Total Non Current Assets	1,838,739	1,681,105	1,520,684	1,357,418	1,191,250	1,022,120	849,969	674,734	496,351	314,758
Total Assets	2,175,190	2,057,332	1,943,599	1,833,799	1,727,703	1,979,533	2,241,209	2,513,009	2,795,218	3,088,127
Liabilities										
Current Liabilities										
Paybles	30,705	31,626	32,575	33,552	34,559	35,596	36,663	37,763	38,896	40,063
Remaining amount of Loan	246,753	270,146	295,755	323,793	0-	0-	0-	0-	0-	0
Total current Liabilities	277,458	301,772	328,330	357,345	34,559	35,596	36,663	37,763	38,896	40,063

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non Current Liabilities										
Long Terms Loans	889,694	619,548	323,793	0	0	0	0	0	0	0
Total Long Term Liabilities	889,694	619,548	323,793	0	0	0	0	0	0	0
Total Liabilities	1,167,152	921,320	652,123	357,345	34,559	35,596	36,663	37,763	38,896	40,063
Owners Equity										
Shareholders Contributions	907,889	907,889	907,889	907,889	907,889	907,889	907,889	907,889	907,889	907,889
Statutory Reseve	10,015	22,812	38,359	56,856	78,526	103,605	129,666	156,736	184,843	214,017
Retained Profits	90,134	205,311	345,228	511,708	706,730	932,443	1,166,991	1,410,621	1,663,589	1,926,157
Total Equity	1,008,038	1,136,012	1,291,476	1,476,454	1,693,144	1,943,937	2,204,546	2,475,246	2,756,322	3,048,064
Total Liabilities and Equity	2,175,190	2,057,332	1,943,599	1,833,799	1,727,703	1,979,533	2,241,209	2,513,009	2,795,218	3,088,127

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 25 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	-2,269,723	405,129	392,885	399,756	406,490	413,053	419,406	432,228	445,388	458,894	472,754	7,280,549
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash Flows	-2,269,723	298,212	261,121	239,891	220,249	202,076	185,262	172,389	160,391	149,210	138,792	2,137,433

Table below (42) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 26 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-2,269,723	405,129	392,885	399,756	406,490	413,053	419,406	432,228	445,388	458,894	472,754
Project Value Returned	2,269,723	1,864,595	1,471,710	1,071,954	665,464	252,412	-166,994	-599,222	-1,044,610	-1,503,504	-1,976,258
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 27 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	1,895,301
Payback Period	5
Internal Rate of Return	%23.38

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 28 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%23.38	5	10.75%
Revenues declined by 10%	%20.72	6	10.75%
Operating Expenses Increased by 10%	%20.11	6	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 1,895,301 Jordan Dinars considering that the project provides 15 Job Opportunities for the governorate residents.